

Basic Network Sniffer using Python

Introduction

A Network Sniffer is a tool used to monitor and analyze network traffic. It captures data packets flowing through a network and provides information about communication between devices.

Objective

The objective of this project is to develop a simple network packet sniffer using Python and the Scapy library. The tool captures packets and displays useful information such as source IP address, destination IP address, and protocol type.

Tools Used

- Python
- Scapy Library
- Visual Studio Code
- Windows Operating System

Methodology

1. Install the Scapy library.
2. Create a Python script.
3. Capture network packets using the sniff() function.
4. Extract source IP, destination IP, and protocol information.
5. Display packet details on the screen.

Code Explanation

The program uses Scapy's packet-sniffing functionality to monitor network traffic. Whenever a packet is captured, the program checks whether it contains an IP layer and then prints the source IP, destination IP, and protocol information.

Results

The network sniffer successfully captured packets and displayed:

- Source IP Address
- Destination IP Address
- Protocol Number
- Packet Summary

This helped in understanding how data travels across a network.

Conclusion

The project demonstrates the basic working of a network packet sniffer. It provides practical exposure to packet analysis and network monitoring. The tool can be extended further to filter packets, save captured data, and analyze network performance.

References

1. Python Documentation
2. Scapy Documentation
3. Network Security Fundamentals